

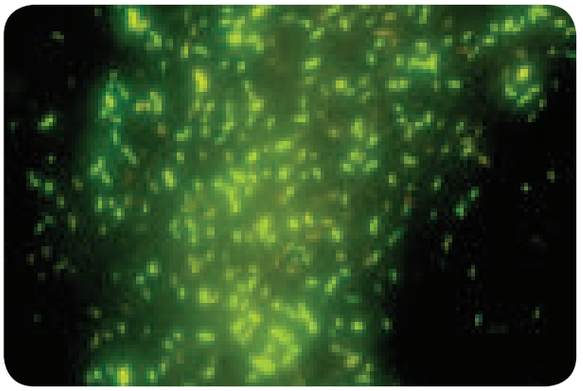


Figure 2.7: **(a)** Fluorescence microscope **(b)** Fluorescent stained tubercule bacilli

Infobits

The Two Types of Fluorescence Microscopes includes diascopic fluorescene and episcopic fluorescene.

Diascopic Fluorescence: K. Reichert and O. Heimstadt demonstrated a fluorescence microscope using auto fluorescent specimens in 1911.

This first type of fluorescence microscopy used transmitted light. Light from the illumination source first passes through an excitation filter and subsequently to the specimen through a dark field condenser. This eliminates most of the excitation light from the imaging side of the system.

Episcopic Fluorescence: In episcopic fluorescence microscopy, the excitation light comes from above the specimen through the objective lens. This is the most common form of fluorescence microscopy today. In this microscope, objective lens acts as both condenser and objective. Quartz objective lenses are required for deep ultraviolet excitation.

2.2.4 Application

- Fluorescence microscope has become one of the most powerful techniques in biomedical research and clinical pathology.
- Fluorescence microscope allows the use of multicolour staining, labeling of structures within cells, and the measurement of the physiological state of a cell.
- Fluorescence microscope helps in observing texture and structure of coal.
- To study of porosity in ceramics, using a fluorescent dye.
- To identify the Mycobacterium tuberculosis.

2.3 Electron Microscope

Examining the ultra structure of cellular components such as nucleus, plasma membrane, mitochondria and others requires 10,000X plus magnification which was just not possible using Light Microscopes. This is achieved by Electron microscopes which have greater resolving power than light microscopes and can obtain higher magnifications.

In an electron microscope, a focused electron beam is used instead of light to

TN_GOVT_XII_Micro_Biology_CH02.indd 19 27-02-2019 13:44:13